Deep Learning (23AID211)

Transfer Learning

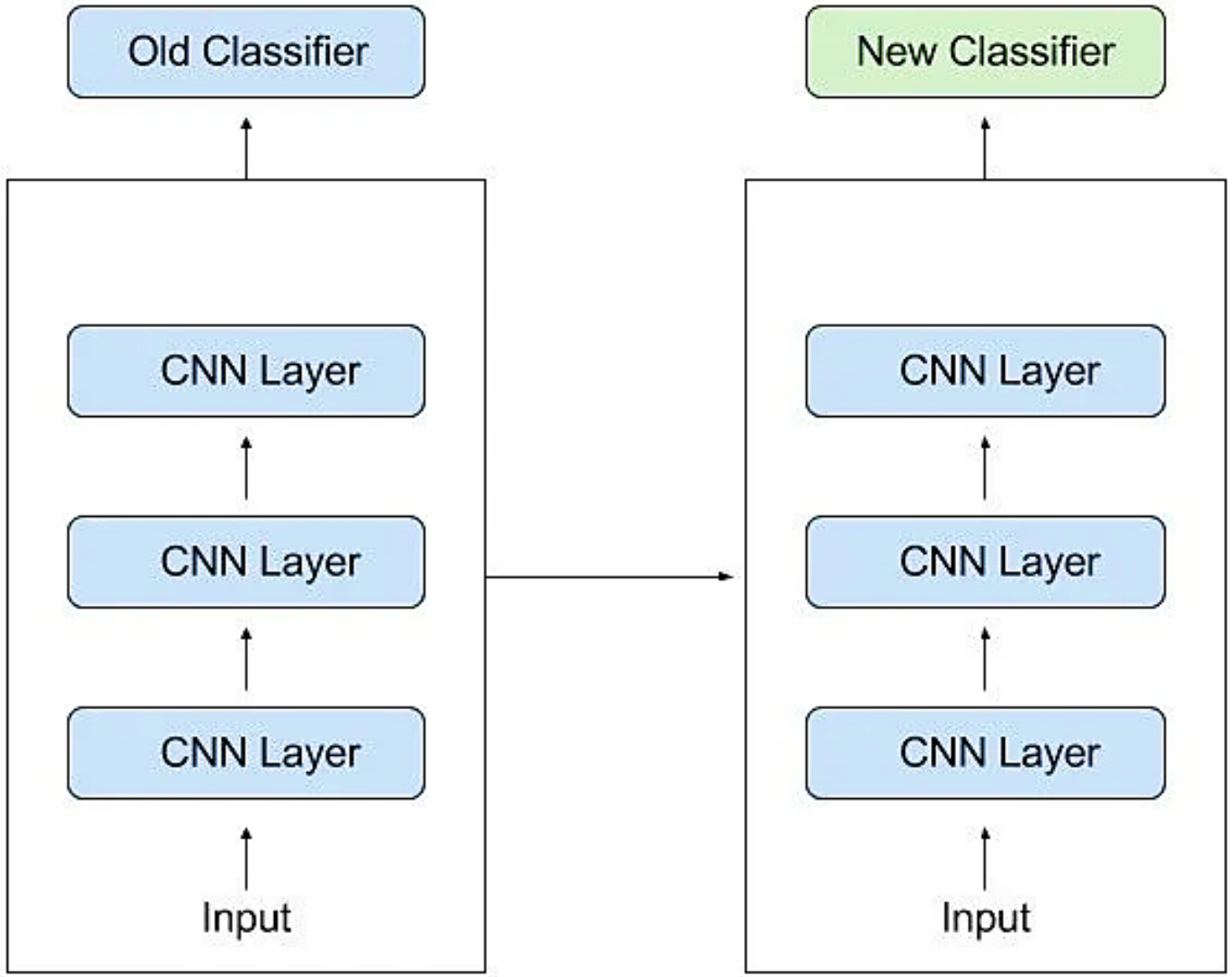
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Transfer Learning

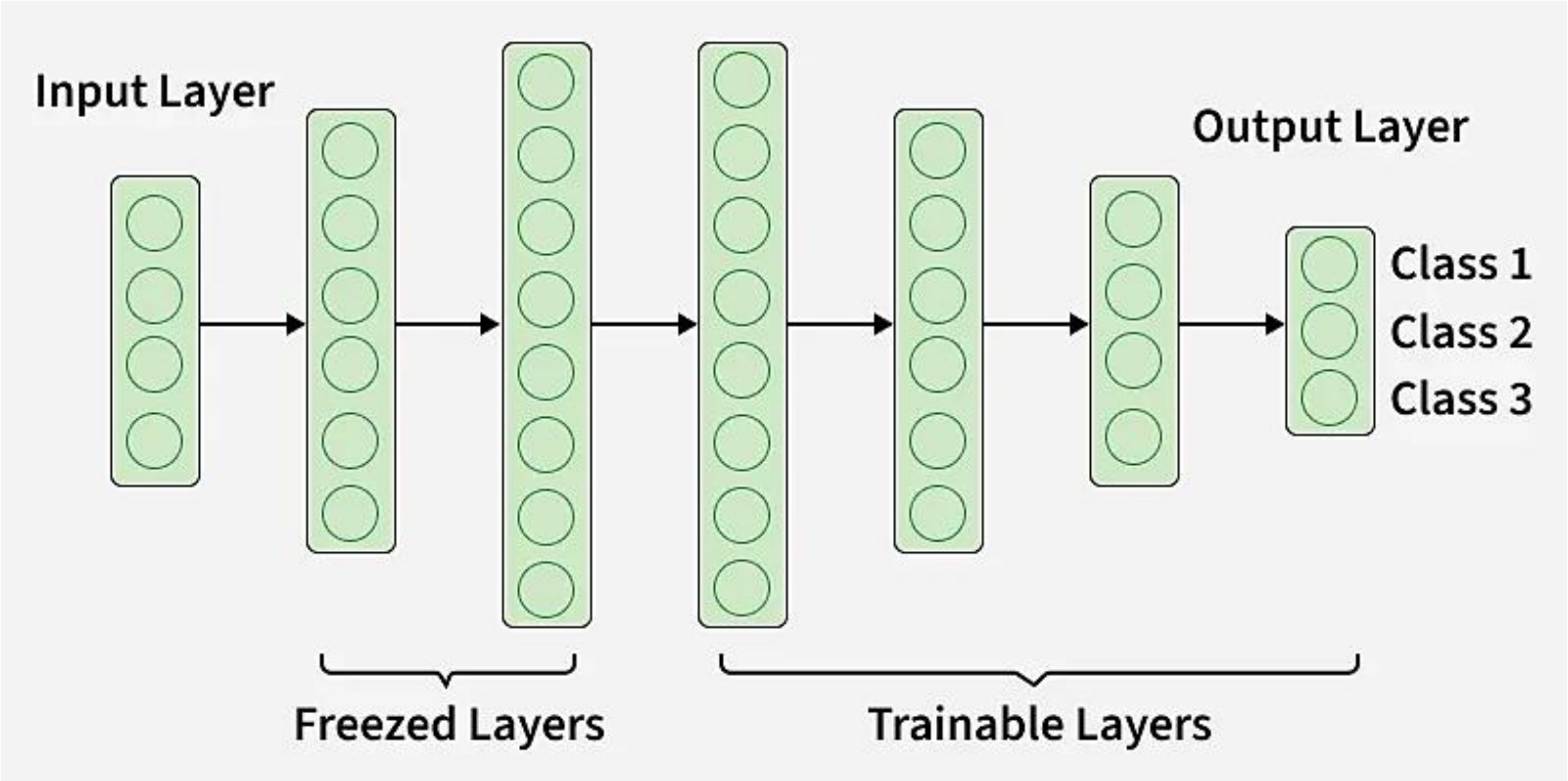
Transfer learning is a technique where a model trained on one task is reused for a related task, especially when the new task has limited data.

- Uses learned features from the first task
- Reduces training time for the new task
- Improves accuracy with less data
- Uses general features that work across tasks

Transfer Learning



Transfer Learning



Transfer Learning

Transfer learning involves a structured process to use existing knowledge from a pre-trained model for new tasks:

Pre-trained Model: Start with a model already trained on a large dataset for a specific task. This pre-trained model has learned general features and patterns that are relevant across related tasks.

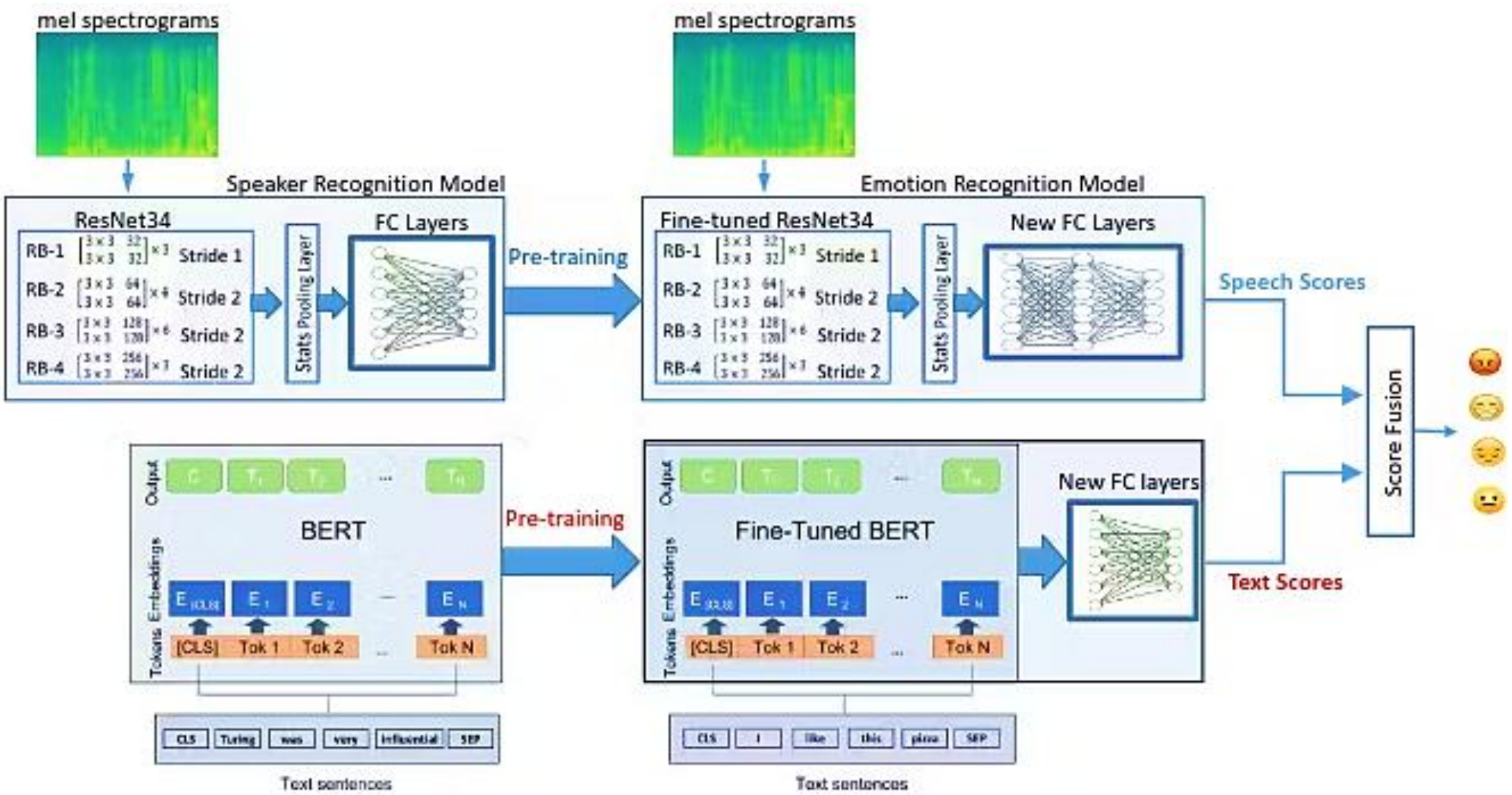
Base Model: This pre-trained model includes layers that process data to learn hierarchical representations, capturing features ranging from low-level to complex.

Transfer Learning

Transfer Layers: Identify layers within the base model that hold generic information applicable to both the original and new tasks. Lower layers capture general features such as edges and textures, while higher layers capture task-specific complex patterns.

Fine-tuning: Fine-tune these selected layers with data from the new task. This process helps retain the pre-trained knowledge while adjusting parameters to meet the specific requirements of the new task, improving accuracy and adaptability.

Transfer Learning



Ethical & Fair Transfer Learning

- The ethical dimension of transfer learning becomes increasingly important as models become more integral to decision-making.
- Future trends will focus on developing fair and unbiased transfer learning methods, ensuring that models don't perpetuate biases in the source data.
- Techniques that enable models to adapt while preserving fairness and avoiding discrimination will be crucial in building AI systems that are ethical, transparent, and accountable.

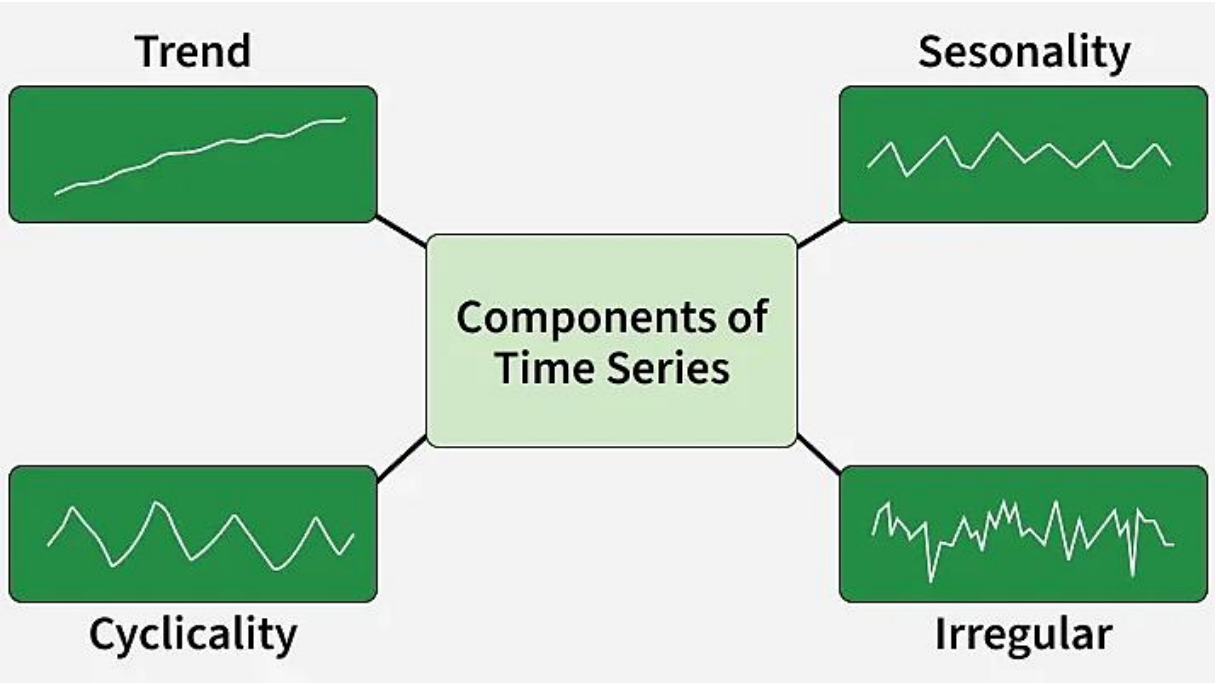
Time Series Analysis

Time Series Analysis

Time series analysis is a statistical method for analyzing a sequence of data points recorded at consistent, chronological intervals to identify patterns, trends, and seasonal variations.

Key Components & ConceptsTrend:

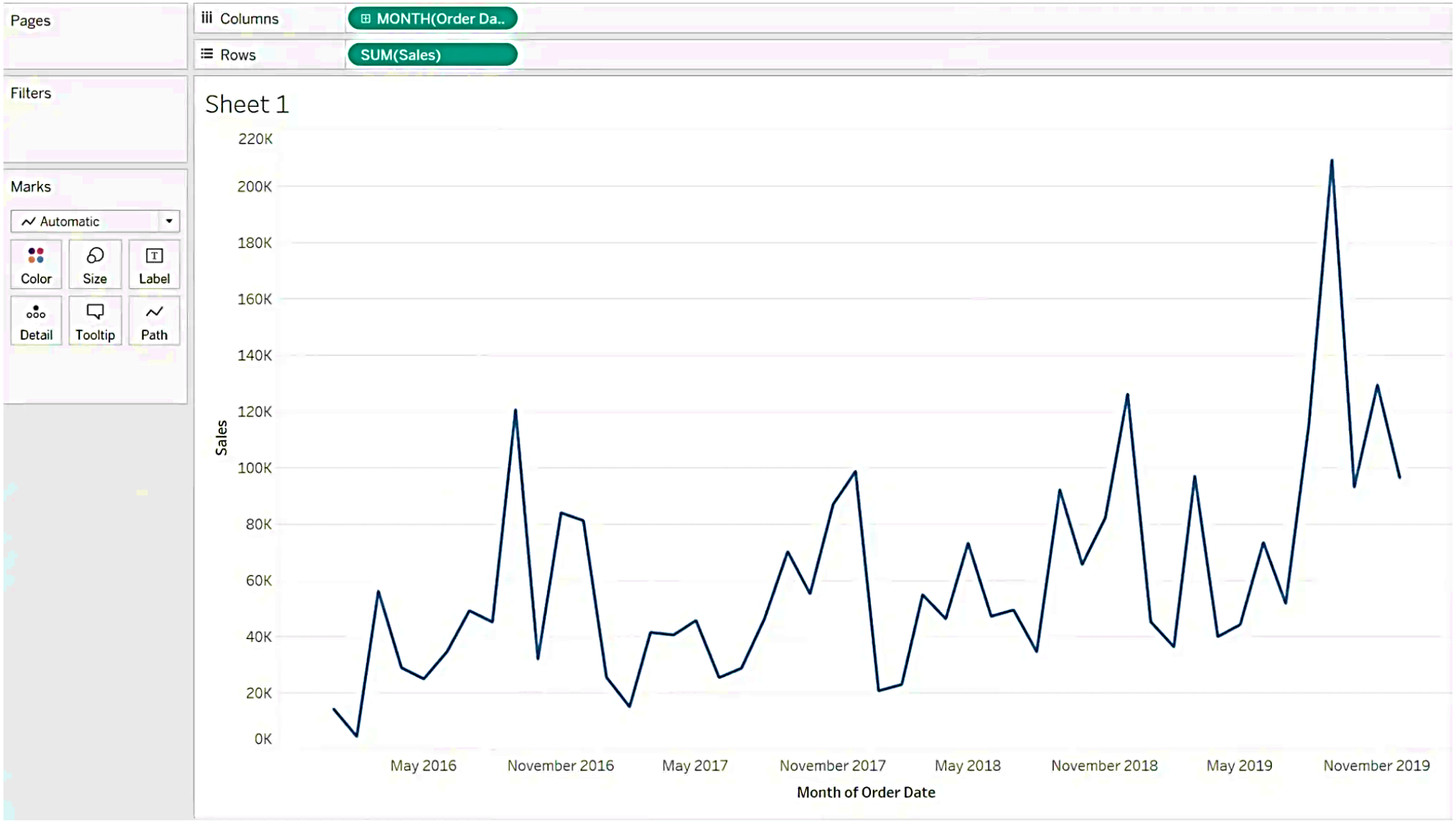
The long-term direction of the data (upward, downward, or flat).



Time Series Analysis

- **Seasonality:** Regular, repeating patterns over fixed periods (e.g., daily, weekly, monthly).
- **Cyclicality:** Trends with no fixed period, often related to economic cycles.Noise/Residuals: Random variations that cannot be explained by the model.
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Time Series Analysis



Time Series Analysis

Time series analysis is used for non-stationary data—data that fluctuate over time or are affected by time. Industries such as finance, retail, and economics frequently use time series analysis because currency and sales are constantly changing. Stock market analysis is an excellent example of time-series analysis in action, especially when applied to automated trading algorithms.

Examples of time series analysis in action include:

Weather data, Rainfall measurements, Temperature readings, Heart rate monitoring (EKG), Brain monitoring (EEG), Quarterly sales, Stock prices, Automated stock trading, Industry forecasts, Interest rates

Models of time series analysis include:

- **Classification:** Identifies and assigns categories to the data.
- **Curve fitting:** Plots the data along a curve to study the relationships of variables within the data.
- **Descriptive analysis:** Identifies patterns in time series data, like trends, cycles, or seasonal variation.
- **Explanative analysis:** Attempts to understand the data and the relationships within it, as well as cause and effect.
- **Exploratory analysis:** Highlights the main characteristics of the time series data, usually in a visual format.
- **Forecasting:** Predicts future data. This type is based on historical trends. It uses historical data as a model for future data, predicting scenarios that could occur at future plot points.
- **Intervention analysis:** Studies how an event can change the data.
- **Segmentation:** Splits the data into segments to show the underlying properties of the source information.